Early Diabetes Risk Screening: Technical Report

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Abstract

Self-reported survey answers can find most adults who have diabetes or prediabetes, at the cost of flagging about three people for every case found. We used 253,680 respondents to the CDC's 2015 Behavioral Risk Factor Surveillance System (BRFSS) survey. We tuned three models on 80% of them, locked a screening cutoff on training data alone, and then scored the held-out 20% once.

At the locked cutoff, XGBoost found **79.4%** of cases (95% CI 78.5–80.3%) while flagging 37.6% of respondents; ROC-AUC was 0.826 and PR-AUC 0.465. Compared with the NIDDK/ADA paper risk test, it found the same share of cases with about **96 fewer false alarms per 1,000 people** screened, and its predicted risks were well calibrated (slope 0.99).

It found 81% of diagnosed diabetes but only 65% of prediabetes, and 44% of cases under age 45. With prediabetes as the only outcome the signal is weak (PR-AUC 0.062 against a 2.1% base rate). The labels are self-reported and cross-sectional, so the model is a research prototype for risk stratification, not a diagnostic tool.

1. Introduction

The question: can self-reported health indicators identify adults at elevated diabetes risk while keeping missed positive cases low? A questionnaire costs almost nothing, so if it can pick out who should get a blood test, screening can reach people who would otherwise not be tested.

We treat missed cases as the main cost. The operating point therefore has to catch a fixed share of cases (80%), and models are compared by how many people they must flag to get there. The bar to clear is the paper Diabetes Risk Test that clinics already use, not just random guessing.

The work follows an eight-stage roadmap: ingestion and quality control, cleaning, exploration, association tests, a leakage-safe split, model tuning, a locked threshold, and reporting. Three decisions were written into the code before the results they govern were seen: the model selection rule (stage 6), the threshold rule and the calibration rule (stage 7). The held-out test set was opened once, after the threshold was saved. Sections 2 to 10 follow the roadmap's stages 1 to 7 in order, and each heading names its stage.

2. Stage 1 · Data ingestion and quality control

The main outcome is **prediabetes or diabetes versus neither**: 39,977 of 253,680 respondents (15.76%) are positive. The data are the three-class file `diabetes_012_health_indicators_BRFSS2015.csv` from the Kaggle Diabetes Health Indicators dataset, derived from the CDC's 2015 BRFSS telephone survey. Each row is one adult with 21 self-reported answers and a diabetes status.

Original status	Respondents	Share	Main endpoint	Sensitivity analysis
0: no diabetes	213,703	84.24%	negative	negative
1: prediabetes	4,631	1.83%	positive	positive
2: diabetes	35,346	13.93%	positive	excluded

The 21 answers are 14 yes/no items (for example high blood pressure, high cholesterol, smoking, physical activity, health coverage, sex), BMI, days of poor physical and mental health in the past 30, and four ordered bands: general health (1 excellent to 5 poor), age (13 five-year bands from 18), education and household income.

Why prediabetes counts as positive. It is the state a screening tool exists to catch. The roadmap's figure of "about 14%" is diabetes alone (13.93%); a row-by-row comparison showed that Kaggle's separate binary file codes all 4,631 prediabetes respondents as negative. The **sensitivity analysis** drops people who already have diabetes and asks the harder question directly: 218,334 respondents, 4,631 with prediabetes (2.12%).

The archive's SHA-256 is `ecbd6572…` and the CSV's `a971809d…`; every later command checks the prepared file against a provenance record of these hashes and counts. The Kaggle and roadmap descriptions say 2015 while UCI's landing page mentions 2014; we report the discrepancy rather than resolve it. The extract has no survey weights, strata or clusters, so all rates describe this table, not the US adult population.

Checks run at load time. The exact shape (253,680 rows × 22 columns), column names and order, integer data types, every value inside its codebook range, and the class labels. The results are saved in `quality.json` and `source_provenance.json`; every later command refuses a file whose hash or counts differ.

3. Stage 2 · Preprocessing and cleaning

Nothing had to be imputed or removed: the file has exactly 253,680 rows and 22 columns, no missing values and no codes outside the codebook. The pipeline is built to stop with an error on such values rather than guess, so a different file cannot slip through silently.

Repeated answers are kept. 23,968 rows repeat another respondent's exact answers. With no respondent ID there is no way to show they are accidental copies, and with 21 coarse answers different people often answer identically: 1,804 repeated profiles include both outcomes. Stage 5 tests whether they inflate model scores (they don't).

Odd answers are reported, not changed. These are valid codes that deserve a second look:

Review flag	Respondents
Told cholesterol is high, but no cholesterol check in five years	1,981
BMI below 15	68
BMI 50 or above	2,547
BMI 60 or above	868
30 of 30 days of poor mental health	12,088
30 of 30 days of poor physical health	19,400

Binary versus ordinal. The 21 answers are 14 yes/no items coded 0/1, four ordered bands (general health 1–5, age 1–13, education 1–6, income 1–8) and three counts (BMI 12–98, and days of poor mental and physical health, 0–30). The bands keep their original codes; the models later treat them either as ordered numbers or as one column per level.

BMI and health-day distributions. Both health-day counts pile up at 0 and at the top code of 30, and BMI has a long right tail; the values are valid, so they are kept and reported rather than clipped.

Variable	Median	Middle 50%	99th percentile	Maximum
BMI (kg/m²)	27	24–31	50	98
Poor mental-health days	0	0–2	30	30
Poor physical-health days	0	0–3	30	30

No scaling, clipping or class balancing happens at this stage. Those steps live inside the model pipelines in stage 5, where they are learned from training folds only and cannot leak information from the test set.

4. Stage 3 · Exploratory data analysis

About one adult in six is positive (39,977 of 253,680, 15.76%), so accuracy would mislead: answering "no" for everyone already scores 84%. That is why the later stages use PR-AUC, recall and precision.

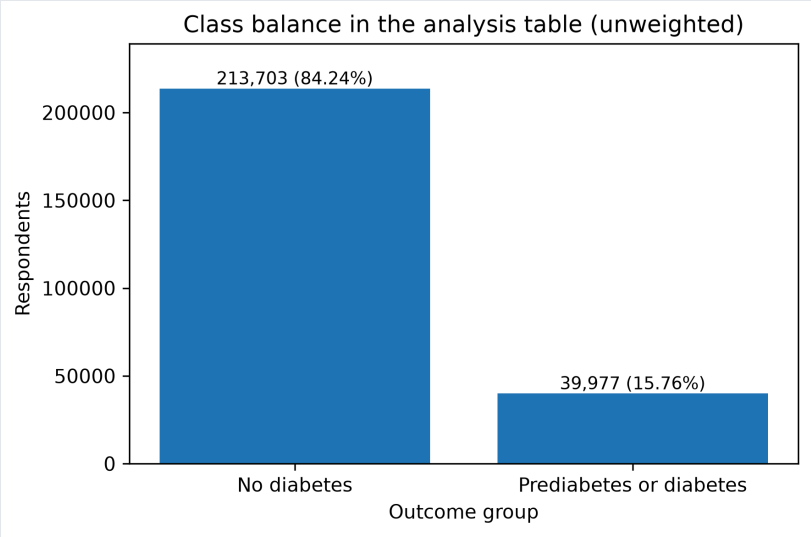


Figure 1. Class balance in the analysis table. Source: reports/uploaded-combined/eda/class_balance.png.

People with diabetes or prediabetes differ most in general health, blood pressure, BMI, walking and cholesterol. Their median BMI is 31 against 27, their median age band 65–69 against 55–59, and their median general health “good” against “very good”. The strongest Spearman correlations with the outcome are general health (0.30), high blood pressure (0.27), BMI (0.24), difficulty walking (0.22) and high cholesterol (0.21).

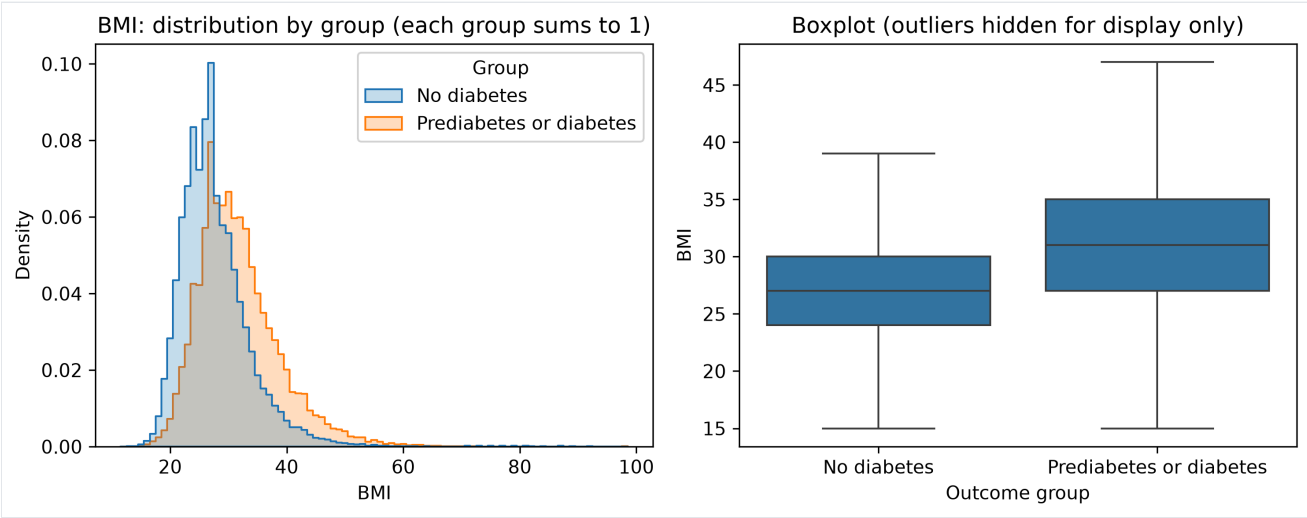


Figure 2. BMI by outcome group, as a histogram and a boxplot (outliers hidden in the boxplot only). Source: reports/uploaded-combined/eda/distribution_BMI.png; age, general health and the health-day counts have matching plots.

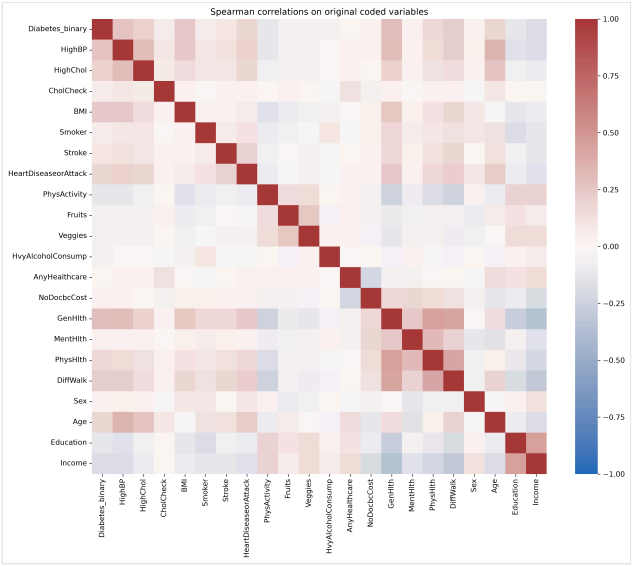


Figure 3. Spearman correlations between the outcome and all 21 answers. Source: reports/uploaded-combined/eda/correlations_spearman.png; the Pearson version is alongside it.

Several of the strongest answers also move together (general health with physical-health days and difficulty walking, age with blood pressure and cholesterol). That is possible confounding, so single-answer comparisons can overstate effects; stage 6 adjusts for it.

5. Stage 4 · Hypotheses and statistics

The tests put numbers on these differences as associations, not causes. Each of the 25 planned tests has a stated null and alternative hypothesis, a statistic, a raw and a Benjamini–Hochberg-adjusted p-value, an effect size, a 95% confidence interval and a one-line interpretation (`hypotheses.csv`).

All 25 planned tests (18 chi-square, 7 Mann–Whitney U) reject independence after Benjamini–Hochberg correction. With a quarter of a million respondents that says little on its own, so the effect sizes carry the result.

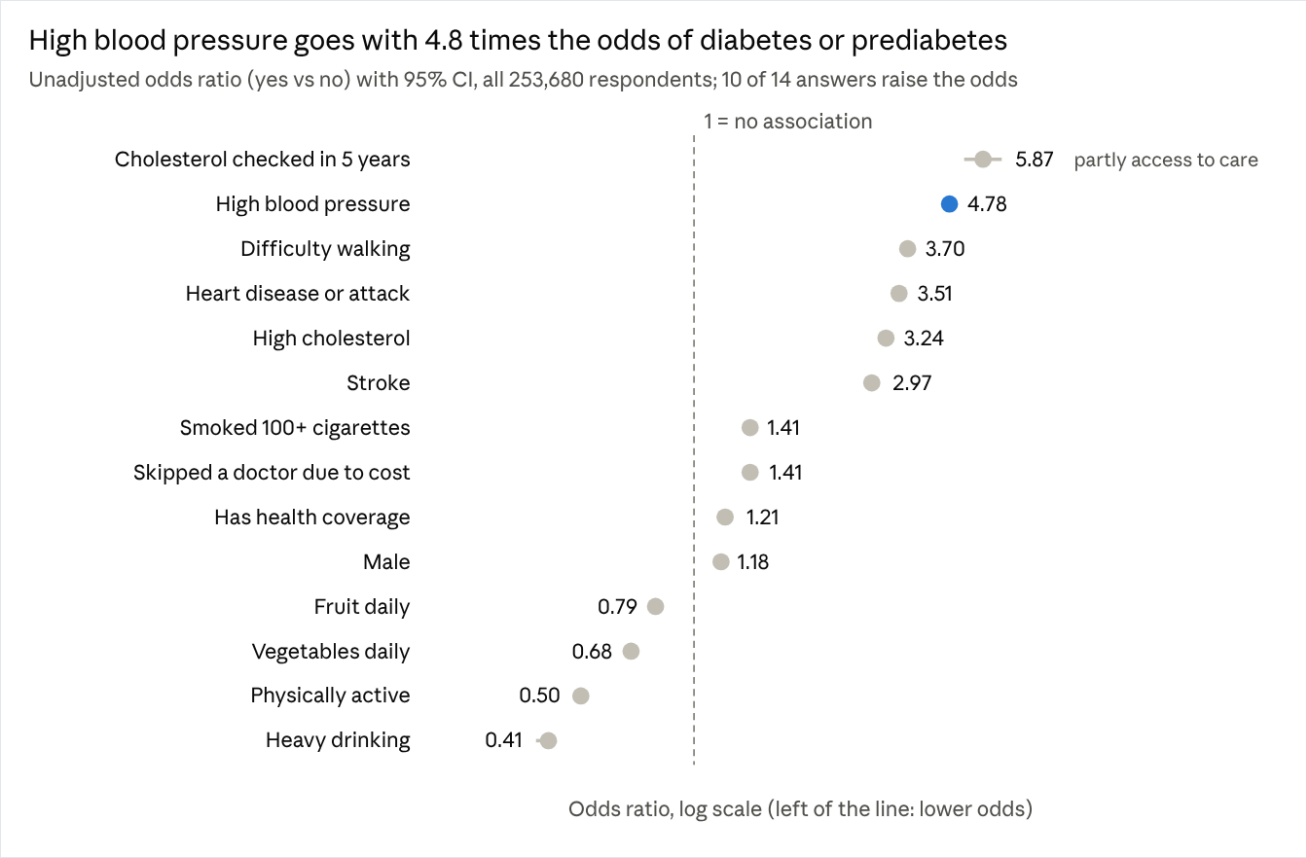


Figure 4. Unadjusted odds ratios for the 14 yes/no answers (chi-square tests). Source: reports/uploaded-combined/hypotheses.csv.

For BMI, the health-day counts and the four ordered bands, Mann–Whitney U tests compare the two groups' distributions; the effect is the rank-biserial correlation (positive means higher values among people with the condition):

Answer	Rank-biserial r (95% CI)	Odds ratio per unit (95% CI)
General health (per step toward poor)	0.450 (0.445–0.455)	2.17 (2.14–2.19)
BMI (per kg/m²)	0.372 (0.366–0.377)	1.083 (1.081–1.085)
Age band (per band)	0.293 (0.288–0.298)	1.21 (1.20–1.21)
Physical-health days (per day)	0.221 (0.215–0.226)	1.045 (1.044–1.046)
Mental-health days (per day)	0.059 (0.053–0.064)	1.024 (1.023–1.025)
Education band (per band)	−0.189 (−0.195 to −0.183)	0.71 (0.70–0.72)
Income band (per band)	−0.266 (−0.271 to −0.260)	0.81 (0.81–0.82)

Two cautions shape the later stages. The largest odds ratio belongs to having had a cholesterol check, which mostly measures contact with doctors, who make the diagnosis. And these are one-answer-at-a-time associations: section 7 (stage 6) shows several shrink or reverse once the other answers are held fixed.

Multiple-testing limitations. Benjamini–Hochberg controls the false discovery rate across the 25 planned tests, but the tests are correlated and further exploratory comparisons were made, so the correction is approximate. None of the intervals is adjusted for multiplicity, confounding or the survey's missing design weights, and p-values this small can underflow to zero.

6. Stage 5 · Splitting and imbalance

One stratified 80/20 split (seed 42) is shared by every later stage: 202,944 training rows (31,982 positive) and 50,736 test rows (7,995 positive). Inside the training part, five fixed stratified folds serve all tuning, model comparison and threshold choice. The

test rows can only be read through a function that refuses them until a `threshold.json` for this exact split exists.

Preprocessing cannot leak. For logistic regression, BMI is robust-scaled, the health-day counts standardised, and the ordered bands either standardised or one-hot encoded (a tuned choice; trees use the raw codes), all inside scikit-learn pipelines, so every scaler is learned from the training part of each fold. A test checks this with an extreme value planted in one validation fold.

Repeated profiles do not inflate scores. 15.0% of rows share their exact answers with someone else. If models were recognising repeated profiles, folds that keep identical profiles together would score lower. They changed validation PR-AUC by at most 0.001 (random forest; 0.0001 for the other two), so the plain stratified split stands.

Class weighting only moves the cutoff. Balanced class weights left PR-AUC unchanged (within 0.001 for all three models) but pushed the average predicted risk from the true 15.8% to about 39%. That is equivalent to lowering the cutoff, and it would make the risks unreadable. The models are therefore unweighted, and stage 7 chooses the cutoff explicitly.

Class weighting against the no-weight baseline, mean of the five validation folds (untuned models):

Model	PR-AUC, no weighting	PR-AUC, balanced weights	Mean risk, no weighting	Mean risk, balanced weights
Logistic regression	0.431	0.430	15.8%	39.5%
Random forest	0.456	0.455	15.8%	38.6%
XGBoost	0.458	0.459	15.7%	38.5%

The true share of positives is 15.8%, so only the unweighted models give risks that can be read at face value.

7. Stage 6 · Training and tuning

XGBoost ranks people best, but the big step is from the paper checklist to any tuned model; the models differ little from each other. Logistic regression (L2), random forest and XGBoost were tuned by randomized search (12, 8 and 16 candidates) on the five folds, scored by PR-AUC. Two yardsticks: no model, and the NIDDK/ADA Diabetes Risk Test scored from the survey answers (age, sex, high blood pressure, inactivity and BMI; 5 or more points means increased risk).

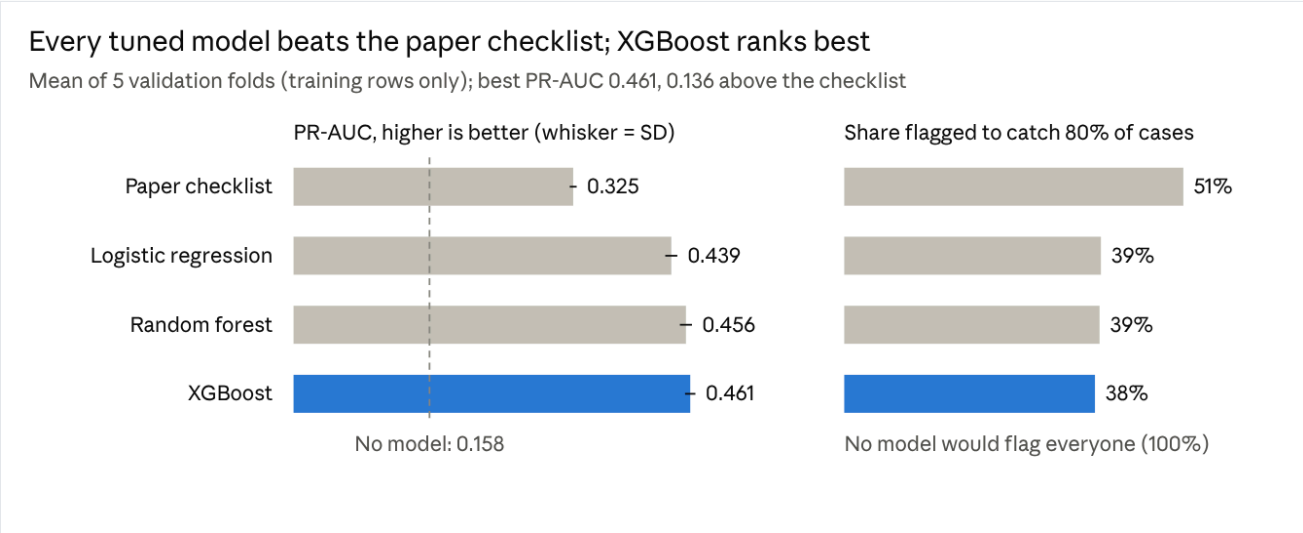


Figure 5. Validation PR-AUC and the share of people each model must flag to catch 80% of cases. Source: reports/modeling/combined/training/comparison.csv.

Selection rule, fixed before the run. Take the model with the best mean PR-AUC; fall back to logistic regression if it is not significantly worse (Nadeau–Bengio corrected resampled t-test, $p < 0.05$); otherwise prefer the lowest Brier score among models tied with the best. XGBoost beat logistic regression in all five folds (+0.022 PR-AUC, $p < 0.001$) and the random forest in all five (+0.005, $p = 0.005$), so it went forward.

Tuned hyperparameters (randomized search over the same five stratified folds, ranked by PR-AUC; the winning settings were then refitted on all training rows and saved as pipelines):

Model	Candidates	Best settings	Validation PR-AUC
Logistic regression (L2)	12	C = 3.47; ordered bands one-hot encoded	0.439
Random forest	8	200 trees, maximum depth 12, at least 10 people per leaf, 33% of answers per split	0.456
XGBoost	**16**	**800 trees, depth 3, learning rate 0.058, minimum child weight 25, row sampling 0.71, column sampling 0.64, L2 penalty 6.8**	**0.461**

What drives the predictions. Permutation importance agrees across all three models: BMI, general health, age, high blood pressure and high cholesterol. Shuffling any other answer lowers validation PR-AUC by 0.006 or less. Refitting without the five healthcare-contact answers costs about 0.03 PR-AUC in every model, which still leaves them far ahead of the checklist.

Adjusted odds ratios (statsmodels, one logistic regression with all 21 answers on the training rows) show how single-answer comparisons mislead:

Answer	Unadjusted OR	Adjusted OR (95% CI)	What changed
Physical-health days, per day	1.045	0.993 (0.992–0.995)	Reversed: carried by general health and walking difficulty
Smoked 100+ cigarettes	1.40	0.98 (0.96–1.01)	Gone: smokers are older and more often men
Vegetables daily	0.69	0.98 (0.94–1.01)	Gone: tracks income and general health
Difficulty walking	3.68	1.10 (1.07–1.14)	Mostly explained by age, weight and health
Physically active	0.50	0.95 (0.92–0.98)	Mostly explained by BMI, health and age
Heavy drinking	0.41	0.52 (0.48–0.56)	Still below 1; likely reverse causation

The strongest adjusted associations are a cholesterol check (3.33), high blood pressure (2.03), high cholesterol (1.81) and each step worse in general health (1.68). No answer has a variance inflation factor above 1.79, so collinearity is not a concern.

8. Stage 7 · Evaluation and thresholds

On the 50,736 held-out people, XGBoost at the locked cutoff caught 79.4% of cases and matched its validation performance, so neither the model nor the cutoff was overfitted.

Choosing the cutoff, on training rows only. The saved XGBoost pipeline was refitted on four of the five folds at a time to give every training row a risk from a model that never saw it. These out-of-fold predictions reproduced stage 6 (PR-AUC 0.4609 against 0.4610). The rule, set before the run, takes the highest cutoff that still flags at least 80% of training cases: **0.1530**. Calibration passed the pre-set rule (slope 0.992, intercept 0.000, within 0.9–1.1 and ±0.1), so no recalibration was applied. The cutoff was saved to `threshold.json` with the hashes of the input, split and model; rerunning the step reproduces that file byte for byte.

What other targets would have cost, on the same out-of-fold predictions (per 1,000 people, about 158 of them with the condition; only the locked row was used):

Rule	Cutoff	Flagged	Cases found	False alarms
Catch 70% of cases	0.209	294	110	184
Catch 75%	0.181	334	118	216
Catch 80% (locked)	**0.153**	**380**	**126**	**254**
Catch 85%	0.122	438	134	304
Catch 90%	0.090	510	142	368
Best F2	0.125	432	133	299

The test set, scored once. Intervals are 95% bootstrap percentile intervals (1,000 resamples of respondents, cutoff held fixed). Logistic regression (cutoff 0.143, chosen by the same rule) and the checklist at its own 5-point line are shown for context; neither

was used to decide anything.

Model	Recall	Precision	Specificity	F2	PR-AUC	ROC-AUC
Paper checklist, 5+ points	0.800 (0.791–0.809)	0.267 (0.261–0.272)	0.589	0.571	0.325	0.756
Logistic regression	0.794 (0.785–0.803)	0.324 (0.317–0.331)	0.690	0.615	0.442	0.819
XGBoost (selected)	**0.794 (0.785–0.803)**	**0.333 (0.326–0.339)**	**0.702 (0.698–0.707)**	**0.622**	**0.465 (0.454–0.476)**	**0.826 (0.821–0.830)**

PR and ROC curves. Across every cutoff, both tuned models stay above the paper checklist on the precision–recall curve and the ROC curve; the dots mark each model's locked operating point.

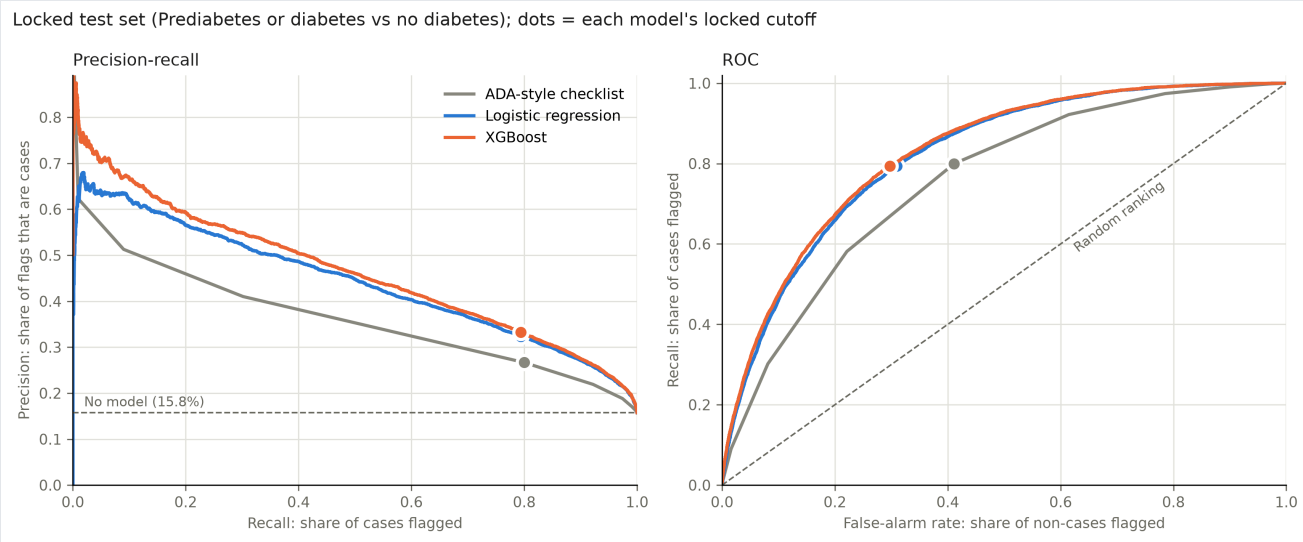


Figure 6. PR and ROC curves on the 50,736 test rows. Source: reports/modeling/combined/evaluation/curves.png.

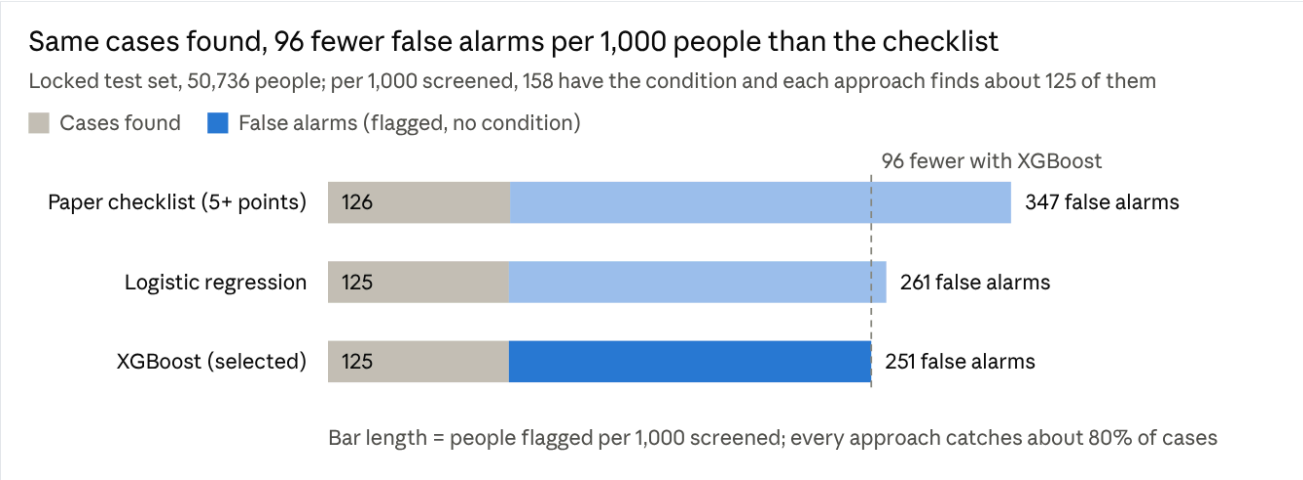


Figure 7. People flagged per 1,000 screened, split into cases found and false alarms. Source: reports/modeling/combined/evaluation/test_results.csv.

Confusion matrix for XGBoost at the locked cutoff: 6,346 cases found, 1,649 missed, 12,721 false alarms and 30,020 people correctly not flagged.

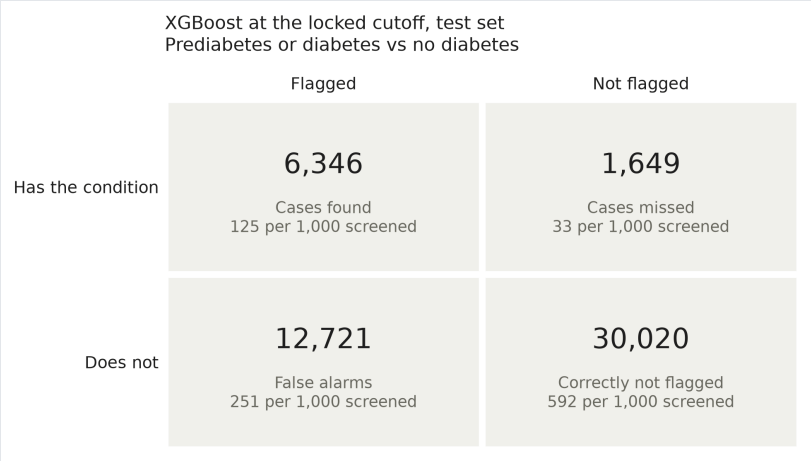


Figure 8. Confusion matrix, counts and rates per 1,000 screened. Source: reports/modeling/combined/evaluation/confusion_matrix.png.

In plain terms, XGBoost sends about 376 of every 1,000 people for a blood test, a third of whom have the condition, and misses about 33. Its risks stay calibrated on new data (observed rate 15.8%, mean predicted 15.7%, slope 0.991, Brier score 0.105), so a stated 30% risk means roughly 30 in 100 such people have the condition. Logistic regression delivers about 90% of the gain over the checklist with a model that can be read line by line.

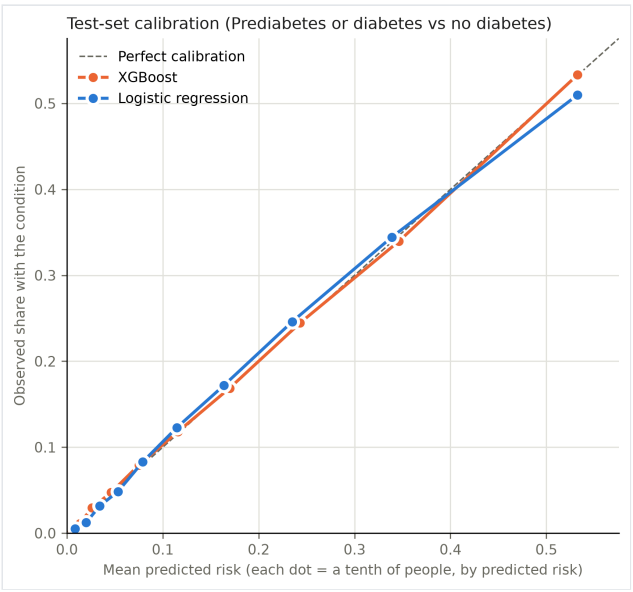


Figure 9. Calibration on the test set: each dot is a tenth of people ranked by predicted risk; the dashed line is perfect calibration. Source: reports/modeling/combined/evaluation/calibration_test.png.

9. Stage 7 · Who the model misses (false negatives)

The 1,649 missed cases look, on paper, like people without the condition, so no cutoff can catch them without flagging far more people. Their average answers sit next to those of non-cases, not of the cases the model catches:

Average answer (test set)	Caught cases (6,346)	Missed cases (1,649)	Non-cases (42,741)
Predicted risk	0.395	0.087	0.125
BMI	32.8	27.5	27.7
Fair or poor general health	48%	9%	13%
High blood pressure	83%	38%	37%
Difficulty walking	43%	9%	13%
Aged under 45	4%	18%	24%

The model leans on BMI, general health, age, blood pressure and cholesterol, so a case with ordinary answers on all of them is invisible to it. Catching 90% instead of 80% would mean flagging about 51% of everyone.

Prediabetes is missed more than diagnosed diabetes. Using the original survey classes, the model flagged 81.3% of people with diabetes (80.3–82.1%) but only 64.7% of people with prediabetes (61.5–67.7%). Prediabetes is 11% of the test cases but 19% of the misses (320 of 1,649). Two likely reasons, which this data cannot separate: a diabetes diagnosis tends to come with diagnosed blood pressure and cholesterol problems and poorer reported health, which the model rewards; and prediabetes is milder, so its answers differ less from everyone else's.

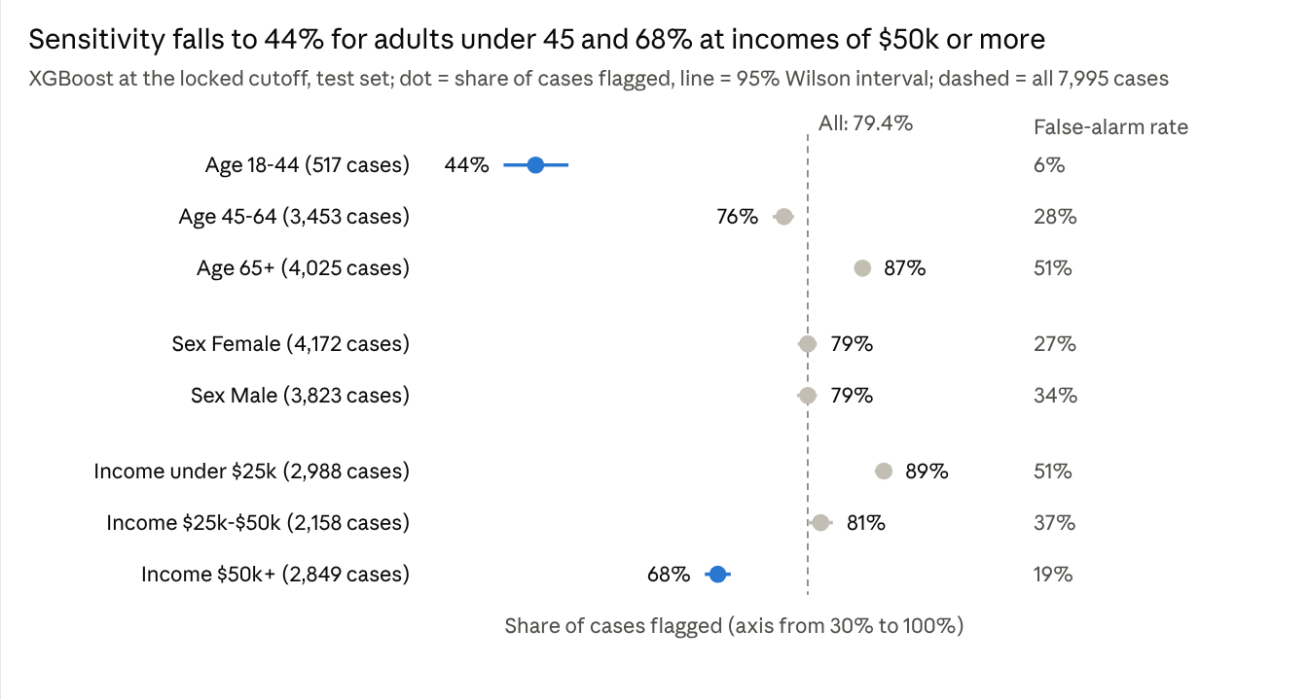


Figure 10. Sensitivity by subgroup with 95% Wilson intervals, and each group's false-alarm rate. Source: reports/modeling/combined/evaluation/subgroups.csv.

One cutoff does not mean one trade-off. For adults under 45 it acts as a cautious test (6% false alarms, 44% of cases found); for people over 65 or with incomes under $25k it flags most cases but also about half of everyone without the condition. Men and women are caught at the same rate, though men get more false alarms. Group-specific cutoffs could even this out but would need to be chosen on training data and justified; we did not tune them.

10. Stage 7 · Prediabetes only (sensitivity analysis)

Finding prediabetes alone is much harder: to catch 81% of cases the model flags 44% of people, and only 4% of those flagged report prediabetes. The same pipeline was run on 218,334 respondents without diagnosed diabetes (2.1% positive), with its own split, tuning, model selection and locked cutoff (0.0172). The test set held 43,667 people, 926 with prediabetes.

Model	Recall (95% CI)	Precision	Share flagged	PR-AUC (95% CI)	ROC-AUC
No model	–	0.021	–	0.021	0.500
Paper checklist, 5+ points	0.702 (0.672–0.731)	0.036	0.418	0.044	0.694
Logistic regression	0.812 (0.786–0.837)	0.039	0.439	0.059	0.757
XGBoost (selected)	**0.814 (0.788–0.840)**	**0.040**	**0.437**	**0.062 (0.056–0.071)**	**0.759**

Per 1,000 people screened (about 21 with prediabetes), XGBoost finds 17 and misses 4; the checklist flags a similar number of people but finds 15 and misses 6. Here the model's gain is in cases found, not in false alarms. Predicted risks stay calibrated (slope 0.998).

The low precision partly reflects the labels. Most people with prediabetes have never been told, so some "false alarms" are probably undiagnosed cases; this survey cannot confirm that. Subgroup gaps are sharper than in the main analysis: recall is 45% for adults under 45 (89 test cases, CI 35–55%) and 92% for those over 65, who then face a 71% false-alarm rate.

11. Limitations

The biggest limit is the label: it records who *reported* diabetes or prediabetes in 2015, so the model identifies current, known status rather than future or hidden risk.

- **Self-reported, cross-sectional outcome.** Nothing here predicts who will develop diabetes. Undiagnosed diabetes and prediabetes sit among the negatives, which lowers measured precision and may teach the model the profile of people who get tested.
- **Diagnosed diabetes is 88% of the positives.** Part of what the model learns is the footprint of an existing diagnosis and its treatment; prediabetes, the case screening exists for, is found less often (65%).
- **Healthcare-contact answers.** Cholesterol checks, high blood pressure and high cholesterol are partly diagnoses that need a doctor. Without the five such answers the models lose about 0.03 PR-AUC but still beat the checklist; a tool built on them would serve people who rarely see a doctor less well.
- **One cutoff, unequal trade-offs.** Sensitivity is 44% under age 45 and 68% at incomes of $50k or more, while over-65s face a 51% false-alarm rate. Group-specific cutoffs were not tuned.
- **Unweighted, single-year extract.** There are no survey weights or design variables, so no population prevalence or subgroup rates should be read as national estimates. Performance on other survey years or on laboratory-confirmed outcomes (such as NHANES) is untested.
- **Associations, not causes.** The adjusted odds ratios are conditional associations in a cross-sectional survey; several show signs of reverse causation (for example heavy drinking appearing protective).

12. Reproducibility and software

Every result in this report rebuilds from the public Kaggle file with one command per stage, and we reran the pipeline on a second operating system to check. Stages 1 to 6 were first run on Linux and stage 7 on Windows (Python 3.13.9), all with the library versions pinned in `requirements.lock` (scikit-learn 1.9.1, XGBoost 3.4.1, statsmodels 0.15.0, pandas 3.0.6, NumPy 2.5.4).

Rerun on Windows	Result
Data preparation	Prepared files byte-identical to the recorded hashes (`7e0e0421…`, `33018d98…`)
Stages 1–4	Every aggregate table and count identical
Split (stage 5)	Split files byte-identical for both endpoints
Model training (stage 6)	Within 0.0003 PR-AUC and 0.001 ROC-AUC of the original run; the same model chosen
Threshold (stage 7)	`threshold.json` byte-identical
Test evaluation (stage 7)	Every table byte-identical

The rerun exposed two portability bugs, both fixed. Prepared files picked up Windows line endings, which changed their hash; they are now always written with Unix line endings. XGBoost models pickled on Linux would not load on Windows; a loader now reads their platform-independent model part without altering the committed files.

The code is a Python package with two commands: `diabetes-screening` for stages 1–4 and `diabetes-screening-model` for stages 5–7. It has 63 automated tests (98% statement coverage), run in continuous integration. Survey rows are never committed: the repository holds aggregate outputs, the fitted logistic regression and XGBoost pipelines, and seven notebooks, one per stage, that read them. Every chart is exported at 300 DPI for print.

References

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2. UCI Machine Learning Repository. CDC Diabetes Health Indicators. A mirror of the binary file.
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Use of AI tools

The code, tests, write-ups and notebooks for stages 5 to 8, and a first draft of this report, were prepared with help from an AI assistant (Claude, by Anthropic) in working sessions with the team members responsible for each stage. Every number in this report comes from the committed outputs in the repository. The decisions that shape the results (the endpoint, the selection rule, the 80% sensitivity target and the calibration rule) were fixed before the results they govern were produced. Each team member has reviewed and run the parts they are responsible for and can explain each step, as the course's generative-AI policy requires.

Before submission: each owner should confirm the last sentence is true, or edit it to describe their own review.